Credit Card Customer Segmentation Using K-Means Clustering

Introduction

This report explores K-Means clustering, an unsupervised machine learning method used to uncover patterns in unlabeled data.[1] It groups similar observations based on their distance to cluster centroids.[2]

Distance between data points is measured relative to cluster centroids, which serve as the centers of each cluster and are initialized before clustering begins. K-Means clustering has many business applications, including customer segmentation, where it is used to identify patterns in customer behavior and inform marketing strategies that can provide a competitive advantage.[3]

This analysis applies K-Means clustering to segment customer credit card usage behavior. The data is first scaled to ensure all variables have comparable influence to the clustering process, reducing bias from differing feature magnitudes. Principal Component Analysis (PCA) is then used to reduce dimensionality while preserving much of the variation in the data. These preprocessing steps reduce the influence of different feature scales and allow the data to be visualized in two dimensions.

Data

The dataset, *Credit Card Dataset for Clustering*, is sourced from Kaggle and contains 8,950 credit card holders over a six month period.This dataset is commonly used for customer segmentation by grouping individuals with similar purchasing and financial behavior patterns using clustering methods.

The dataset contains 18 variables describing customer spending, payment behavior, credit utilization, and account activity. These variables include measures of purchasing patterns,

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[1]Azika Amelia, "K-Means Clustering: From A to Z," *Towards Data Science* (Medium), September 28, 2018, https://towardsdatascience.com/k-means-clustering-from-a-to-z-f6242a314e9a.
[2]Mayank Varia, *DS121: Foundations of Data Science II*, class lecture, Boston University, Boston, September 13, 2022.
[3]Abhinav Sagar, "Customer Segmentation Using K Means Clustering," *Towards Data Science* (Medium), December 3, 2019, https://towardsdatascience.com/customer-segmentation-using-k-means-clustering-d33964f238c3.

account balances, payment activity, credit limits, cash advance usage, and account tenure.[4] Descriptions of each variable are provided in the table below:

Table 1. Data Dictionary

Variable	Description
CUST_ID	Identification of Credit Card holder (Categorical)
BALANCE	Balance amount left in their account to make purchases (
BALANCE_FREQUENCY	How frequently the Balance is updated, score between 0 and 1 (1 = frequently updated, 0 = not frequently updated)
PURCHASES	Amount of purchases made from account
ONEOFF_PURCHASES	Maximum purchase amount done in one-go
INSTALLMENTS_PURCHASES	Amount of purchase done in installment
CASH_ADVANCE	Cash in advance given by the user
PURCHASES_FREQUENCY	How frequently the Purchases are being made, score between 0 and 1 (1 = frequently purchased, 0 = not frequently purchased)
ONEOFF_PURCHASES_FREQUENCY	How frequently Purchases are happening in one-go (1 = frequently purchased, 0 = not frequently purchased)
PURCHASES_INSTALLMENTS_FREQUENCY	How frequently purchases in installments are being done (1 = frequently done, 0 = not frequently done)
CASH_ADVANCE_FREQUENCY	How frequently the cash in advance being paid
CASH_ADVANCE_TRX	Number of Transactions made with "Cash in Advanced"
PURCHASES_TRX	Number of purchase transactions made
CREDIT_LIMIT	Limit of Credit Card for user
PAYMENTS	Amount of Payment done by user
MINIMUM_PAYMENTS	Minimum amount of payments made by user
PRC_FULL_PAYMENT	Percent of full payment paid by user
TENURE	Tenure of credit card service for user

[4]Arjun Bhasin, "Credit Card Dataset for Clustering," *Kaggle*, March 2, 2018, https://www.kaggle.com/datasets/arjunbhasin2013/ccdata.

Methodology

The analysis begins by importing key Python libraries for data processing, visualization, and clustering, including Pandas, NumPy, Matplotlib, PowerTransformer, MinMaxScaler, KMeans, PCA, and evaluation tools such as the Elbow and Silhouette visualizers.

The dataset was inspected and a total of 313 missing values were found in the "MINIMUM_PAYMENTS" variable. Because K-Means requires complete observations, the missing values were examined before clustering. An exploratory review indicated that many missing values in the MINIMUM_PAYMENTS variable corresponded to observations with zero recorded payments. Based on this pattern, the missing values were imputed with zero to preserve the dataset for clustering. Although the exact reason for the missing values is unknown, this provided a reasonable preprocessing approach for this exploratory analysis.

After addressing the missing values, the data was prepared for K-Means clustering. Because K-Means is sensitive to differences in feature scales, the variables were evaluated and transformed to ensure that no single feature disproportionately influenced the clustering results. To determine an appropriate preprocessing approach, the distributions of the variables were examined to assess normality, and a correlation analysis was conducted to identify highly correlated features.

Inspection of the variable distributions revealed that many features were highly skewed and deviated from normality. A power transformation was applied to reduce skewness and stabilize the distributions. The transformed features were then scaled to a 0–1 range to ensure that all variables contributed more equally to the clustering process.[5]

To assess multicollinearity, I generated a correlation matrix for all variables and identified several features with strong positive correlations. Because highly correlated variables can introduce redundancy and influence clustering results, this analysis informed the subsequent use of PCA for dimensionality reduction.

[5]Jason Brownlee, "How to Use Power Transforms for Machine Learning," *Machine Learning Mastery*, August 27, 2020, https://machinelearningmastery.com/power-transforms-with-scikit-learn/.

After transforming and scaling the data, I applied Principal Component Analysis (PCA) to reduce dimensionality and project the data into two principal components for visualization and cluster interpretation. PCA helped capture the variance in the data while reducing redundancy from highly correlated features, making it easier to evaluate and interpret customer segments.

To determine the optimal number of clusters, I evaluated K-Means models using the elbow method, silhouette scores, and inertia values across multiple cluster configurations. By comparing these metrics and visualizing the resulting clusters, I selected the most appropriate segmentation and analyzed customer behavior patterns within each group.[6]

Results

The elbow plot suggested that improvements in inertia began to level off around five clusters, so I further evaluated clustering results with four to eight clusters to assess the stability and quality of the segmentation. This comparison provided additional context for selecting the final clustering model and interpreting customer behavior patterns. **Table 2** summarizes the silhouette scores and inertia values for the evaluated cluster.

Table 2. Silhouette Scores and Inertia Values by K-Means Cluster

Silhouette Scores and Inertia Values by K-Means Cluster		
Clusters	**Silhouette Score**	**Inertia**
3	0.44907172409886914	2693.756348793962
4	0.4717094210683729	1753.6145939784392
5	0.4471183460325966	1440.0786451500398
6	0.38739316519317496	1232.4072855020795
7	0.4349419555453952	1010.7211495665769
8	0.41393487649556765	855.5596791897678

[6] Yellowbrick Documentation, “Elbow Method,” accessed July 6, 2026, https://www.scikit-yb.org/en/latest/api/cluster/elbow.html

Model evaluation revealed a trade off between cluster density and cluster separation. The four-cluster solution achieved the highest silhouette score (0.472), indicating the strongest separation between clusters, while the eight-cluster solution produced the lowest inertia value (856.11), reflecting tighter groupings within clusters. As the number of clusters increased, inertia consistently decreased, while silhouette scores generally declined.

Figure 1. Distortion Score Elbow for KMeans Clustering

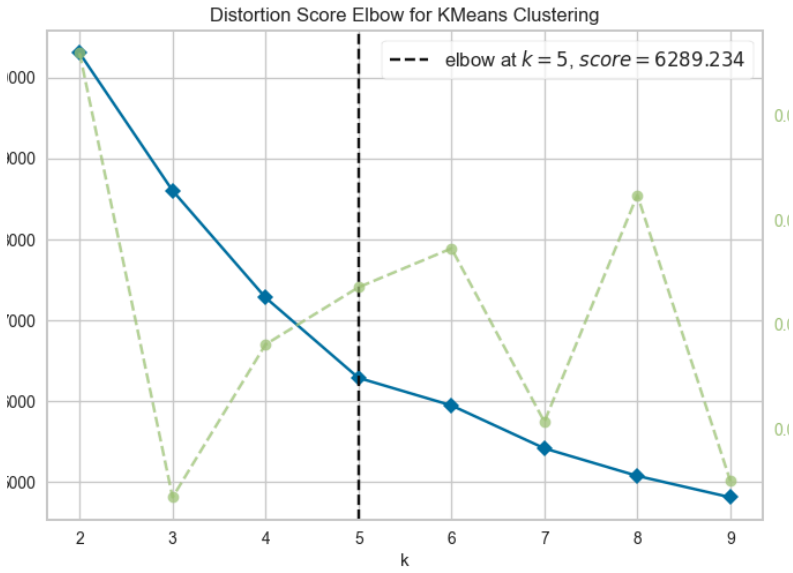
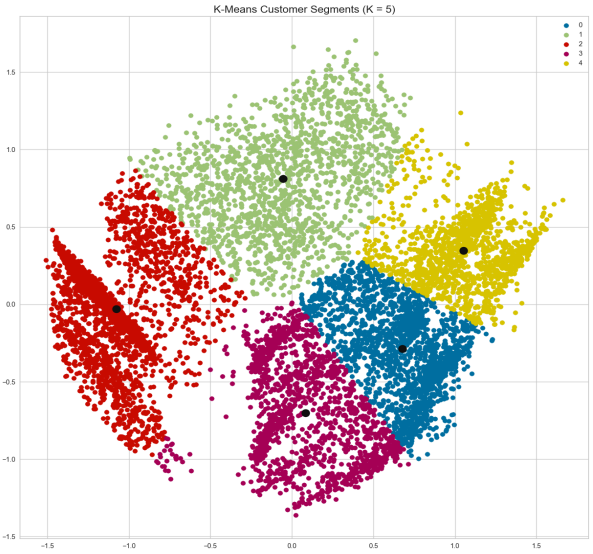


Figure 2. KMeans Clusters (K=5)



The five-cluster solution, selected by the elbow method shown in Figure 1, produced a silhouette score of 0.447 and an inertia value of 1439.98, representing a reasonable balance between cluster compactness and interpretability. Although the four-cluster model achieved a slightly higher silhouette score, the five-cluster solution provided greater segmentation detail while maintaining comparable clustering quality

Using the five-cluster solution as shown in **Figure 2**, I identified several distinct customer spending behaviors. Customers in Cluster 2 exhibited the highest levels of overall spending, single purchase transactions, and installment purchases, suggesting a group of highly active

credit card users. Cluster 1 also showed elevated spending activity but was characterized by a greater reliance on cash advances. In contrast, Clusters 3 and 4 demonstrated relatively low spending activity, infrequent cash advances, and fewer single purchase transactions, indicating more conservative credit card usage patterns.

One notable finding was the relationship between purchase behavior and payment methods. **Figure 3** shows distinct differences in purchasing behavior across the five customer segments. Higher levels of single purchase transactions were generally associated with higher overall spending, particularly among customers in Clusters 1 and 2. **Figure 4** illustrates customer clusters based on cash advance transactions and purchase frequency. Most customer segments exhibited relatively low cash advance activity, while Cluster 0 demonstrated a higher number of cash advance transactions despite lower purchase frequency. This suggests that customers in this segment may use their credit cards differently than the other groups.

Figure 3. One-Off Purchases vs. Purchases

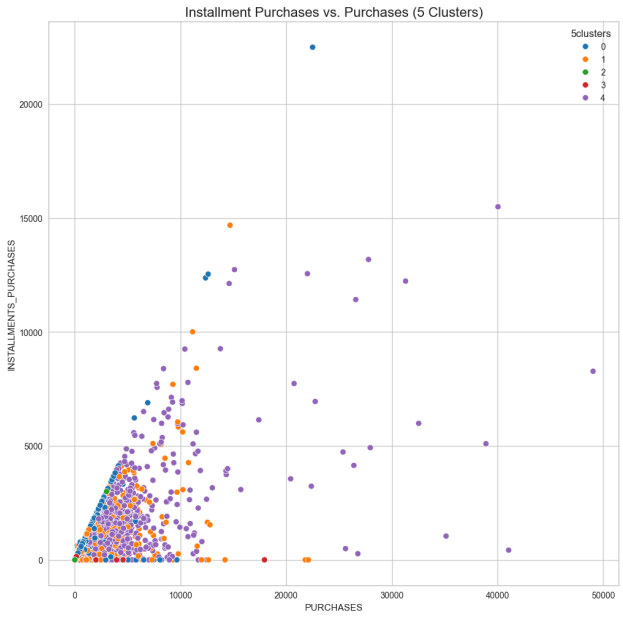
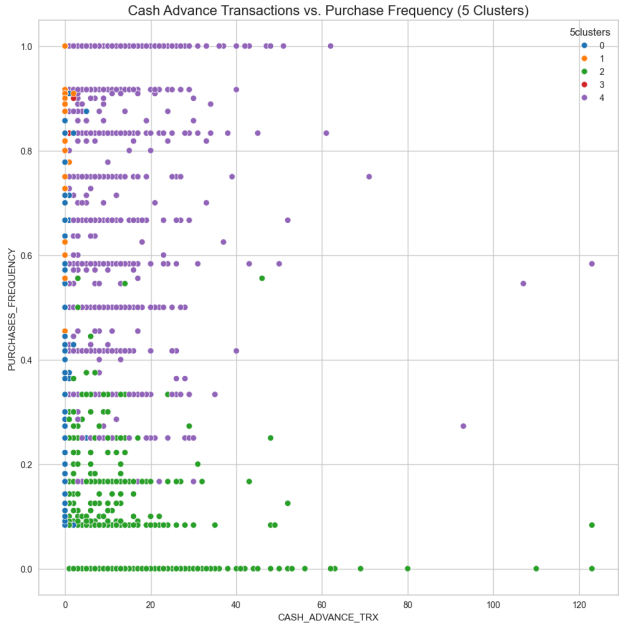


Figure 4. Cash Advance vs. Purchases Frequency



Comparing the five-cluster solution to the four-cluster alternative revealed similar overall patterns, despite the four-cluster model producing a slightly higher silhouette score. This comparison highlighted the importance of evaluating clustering results using multiple metrics rather than relying on a single performance measure.

Conclusion

This project demonstrated the complete workflow for applying K-Means clustering to customer behavior data. After preprocessing the data, transforming and scaling the variables, and reducing dimensionality with PCA, multiple clustering solutions were evaluated using the elbow method, silhouette scores, and inertia values. While the elbow method supported a five-cluster solution, comparing alternative models highlighted the trade-offs between cluster separation and model complexity.

The final segmentation revealed meaningful differences in customer spending behavior, payment habits, and cash advance usage. This analysis has several limitations. The results of K-Means can change depending on how the data is prepared and the number of clusters that is chosen. Because the true number of customer groups is unknown, the five-cluster solution represents one reasonable interpretation of the data.

This project provided practical experience with data preprocessing, dimensionality reduction, cluster evaluation, and interpreting the results of an unsupervised learning model. It also demonstrated how clustering techniques can be used to generate business insights from unlabeled customer data.